Multinomial Processing Tree Models

Block 4.1: Data Science, Lecture 6

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Types of Models

Data-driven Models

- statistical models that are primarily based on the data at hand
- e.g., linear regression, generalized linear models, data simulation, neural networks

Theoretical Models

- psychological-plausible models
- mathematical-formulated theories
- e.g. Theories about decision making or Human memory

Theoretical Models

Typical Procedure

1. Psychological theory
2. Formulate it as “mathematical model”
3. Model fitting and parameter estimation
4. Interpretation

Multinomial Processing Tree (MPT) Models

- statistical models for discrete/categorical data
- clear connection to psychological theory
- conceptual and mathematical simplicity
- easy to estimate parameters from data

Multinomial processing tree models: Applications

Main application area: **Human memory**

- Recognition memory
- Source monitoring
- Storage-retrieval paradigms
- Hindsight bias

But also other areas of cognitive psychology

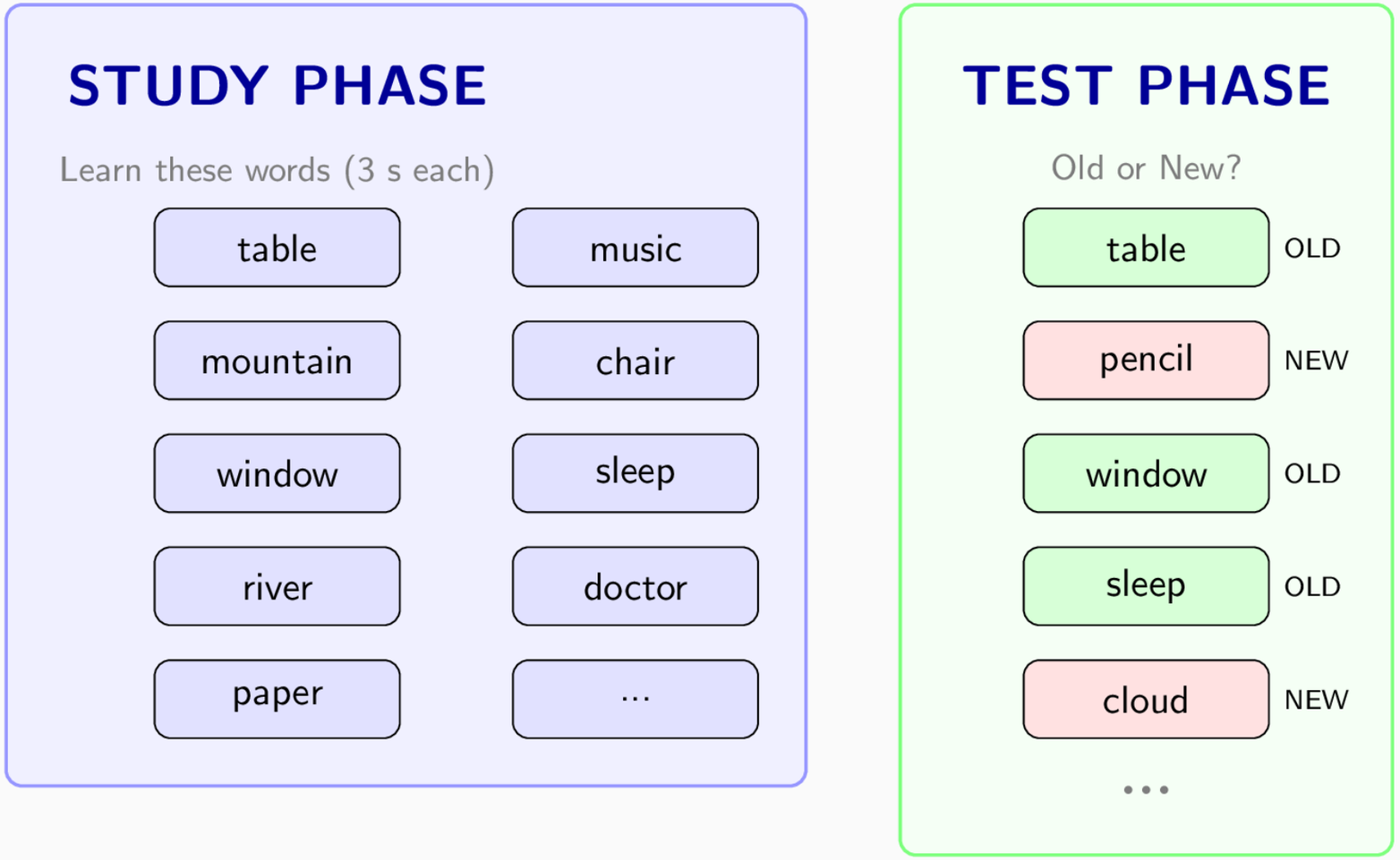
- Perception
- Categorization
- Decision making
- Reasoning

Batchelder & Riefer (1999) and Erdfelder et al. (2009) review applications of multinomial processing tree models in psychology.

Our Memory Model

Example Experiment: Paradigm

Simple recognition memory experiment



Example Experiment: Data

Participant Responses

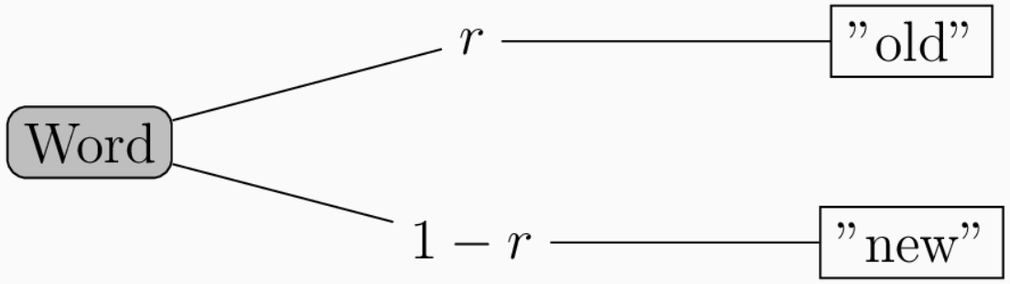
	response: “old”	**response: “new”**
Old word	*Hit*	*Missing*
New word	*False Alarm*	*Correct Rejection*

Example data (3 participants)

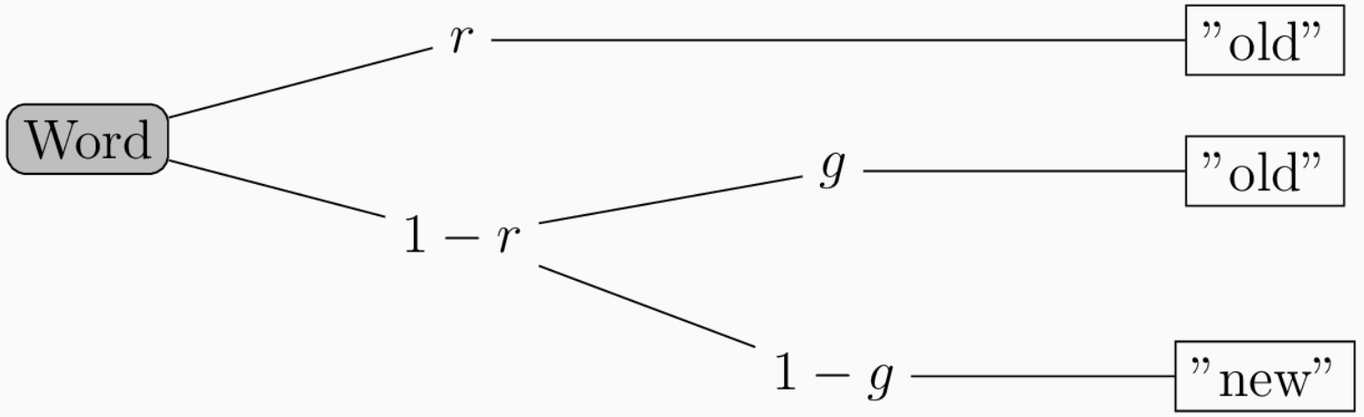
H	**M**	**CR**	**FA**
71	29	17	30
84	25	36	18
62	34	20	35

Example: Simple recognition memory experiment

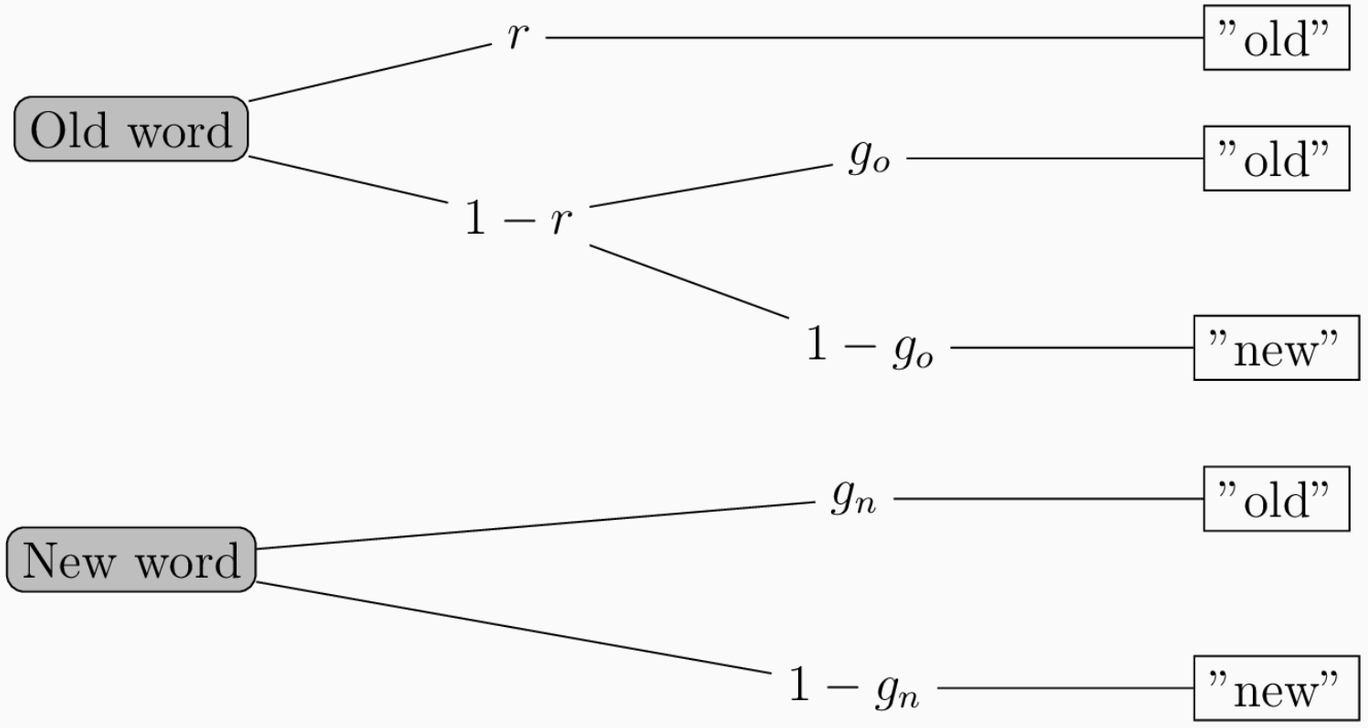
Building a mathematical model of Human Recognition memory



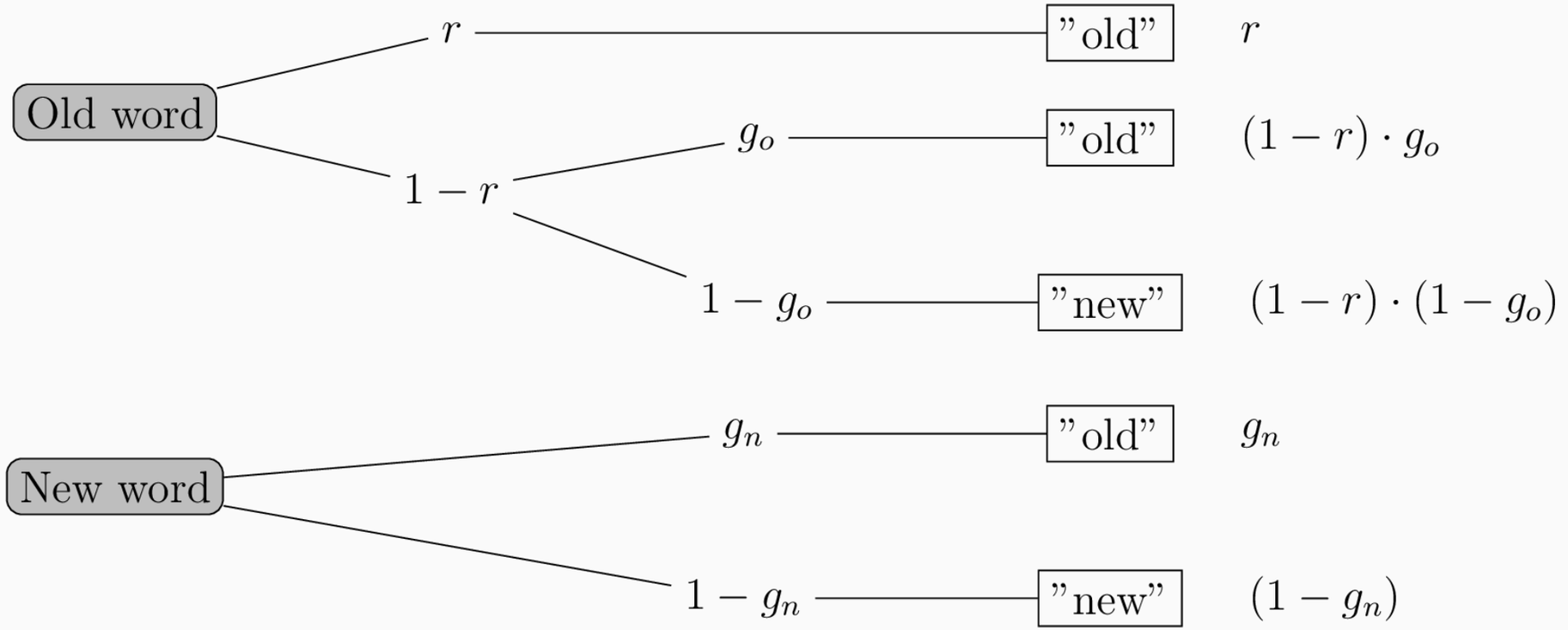
Building a mathematical model of Human Recognition memory



Building a mathematical model of Human Recognition memory

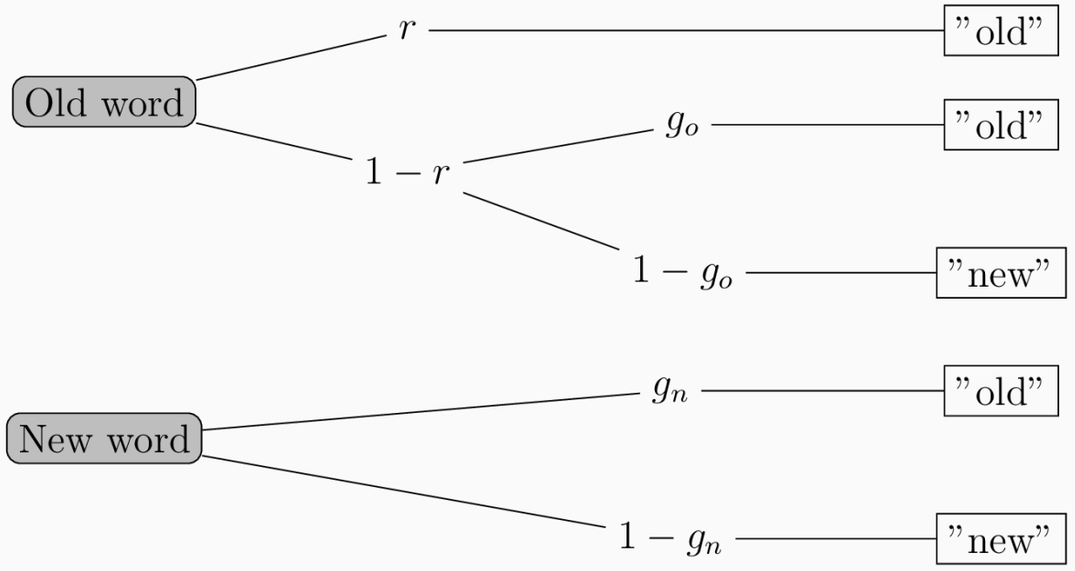


Building a mathematical model of Human Recognition memory



Building a mathematical model of Human Recognition memory

Model: *"One-high-threshold model (1HTM) of recognition memory"*



$$p(\mathrm{H}) = r + (1 - r) \cdot g_o$$
$$p(\mathrm{M}) = (1 - r) \cdot (1 - g_o)$$

$$p(\mathrm{CR}) = 1 - g_n$$
$$p(\mathrm{FA}) = g_n$$

H

Missin

Correct Rejectio

False Alar

Data subject 1

H	M	CR	FA
71	29	17	30

Problem to solve

- Which parameter values best describe to observed data?
- Find the latent parameters r, g_o, and g_n that maximize the likelihood of the data

The Formal MPT Model

Just a little bit math for those how are interested

1. **MPT Model**
   - $\Theta = (\theta_1, \ldots, \theta_S)$: latent model parameters
   - p_j = predicted probability of observing behaviour in category C_j

2. **Data**
   - $D = (N_1, \ldots, N_J)$: observed frequencies in J categories

In our model:

$$D = (H, M, CR, FA)$$
$$\Theta = (r, g_o, g_n)$$

We want to find the parameter values Θ that maximize the likelihood of the observed data D.

1. MPT Model

Just a little bit math for those how are interested

The MPT models defines the probabilities of the i-th branch leading to the j-th category take the form

$$p_{ij}(\Theta) = c_{ij} \prod_{s=1}^{S} \theta_s^{a_{ijs}} (1 - \theta_s)^{b_{ijs}}$$

where

- θ_s is the s-th latent model parameter
- a_{ijs} and b_{ijs} are the number of times θ_s and $1 - \theta_s$ occur in branch i leading to category j.
- c_{ij} is a non-negative real number

Multinomial processing tree models: Likelihood

Just a little bit math for those how are interested

The Likelihood function is the probability of the observed data as a function of the model parameters.

- p_j = probability of observing behaviour in category C_j
- $D = (N_1, \dots, N_j)$: vector of observed frequencies in categories $C_1, \dots, C_j$
- Θ: vector of latent model parameters

Assuming independence of the responses, the data follow a multinomial distribution. The likelihood becomes:

$$L(D|\Theta) = \frac{N!}{\prod_{j=1}^{J} N_j!} \prod_{j=1}^{J} p_j(\Theta)^{N_j}$$

and it depends only on the latent parameters.

Expectation-Maximization (EM) Algorithm

We want to maximize $L(D|\Theta)$. Hu & Batchelder (1994) present a version of the EM algorithm for finding the **maximum likelihood estimates** of MPT model parameters.

The EM is implemented in several R package.

```
1 library(mpt)          # package needs to be installed once via install.packages("mpt")
```

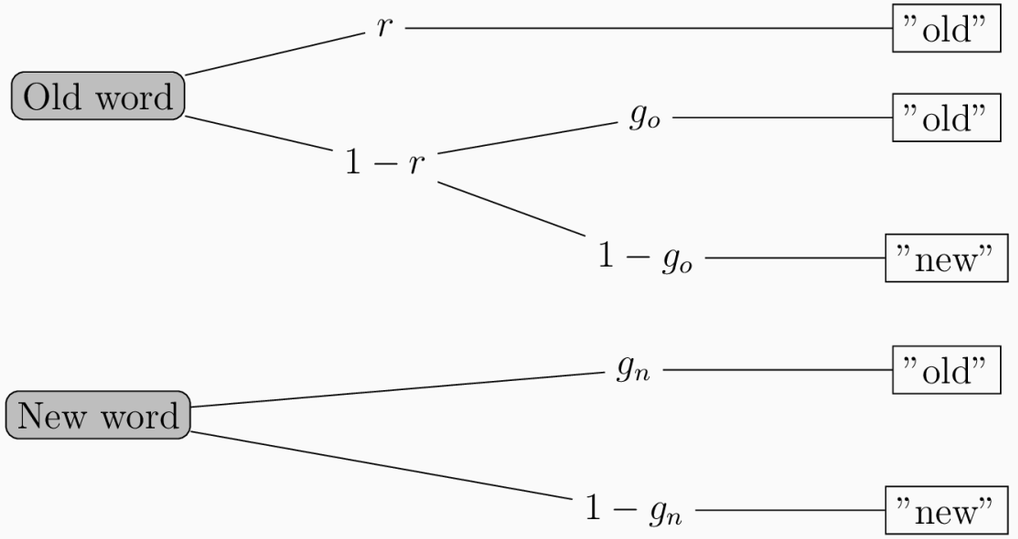
Example

Data

```
dat_text = "
treeid resp freq
    1   H   84
    1   M   25
    2   CR   36
    2   FA   18
"
dat = read.table(text=dat_text, header = TRUE)
```

Model

```r
# define model
md_1htm <- mptspec(
  r + (1-r) * go,
  (1-r)*(1-go),
  1-gn,
  gn
)
```



Example

Fit model

```
1 md_fit_a = mpt(md_1htm, dat)
2 summary(md_fit_a)
```

```
Coefficients:
   Estimate Logit Estim. Std. Error z value Pr(>|z|)
r    0.5211       0.0844        NaN     NaN      NaN
go   0.5211       0.0844        NaN     NaN      NaN
gn   0.3333      -0.6932     0.2887  -2.401   0.0163 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Likelihood ratio G2: 6.661e-15 on -1 df, p-value: NaN
Pearson X2: 3.775e-15,    AIC: 15.137
Number of trees: 2
```

Identifiability

Checking the identifiability of an MPT model

- The number of free parameters (S) **must not** exceed the number of free categories (J) provided by the category system, that is, $S \leq J$.

```
1 nobs(md_fit_a)  ##  J = 2, number of free categories
```

```
[1] 2
```

```
1 length(md_fit_a$coefficients) ## S = 3, number of parameters
```

```
[1] 3
```

Since $S > J$, the model is **not identifiable**. The parameters cannot be estimated uniquely (reliably) from the data.

Example: Restricted Model

$g_o = g_n$

```
1 md2_1htm = update(md_1htm, .restr = list(go=gn))
2 md_fit_b = mpt(md2_1htm, dat)
3 summary(md_fit_b)  ## goodness-of-fit test
```

```
Coefficients:
   Estimate Logit Estim. Std. Error z value Pr(>|z|)
r    0.6560       0.6454     0.3052   2.114   0.0345 *
gn   0.3333      -0.6931     0.2887  -2.401   0.0163 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Likelihood ratio G2: 0 on 0 df, p-value: 1
Pearson X2: 1.343e-15,    AIC: 13.137
Number of trees: 2
```

Model Validation & Comparisons (& Maximum Likelihood Estimation)

Model Validation

Usually, MPT models are used to formalize and test predictions of established psychological theories. That is, we want to test how good is our model.

Models Degrees of freedom

$$df = J - S \qquad (\text{S} = \text{free parameters}, \text{J} = \text{free categories})$$

- $df < 0$: is **not identifiable** ($S \leq J$)
- $df = 0$: means that the model is **saturated**
  - fits the data perfectly, but **cannot** be validated.
- $df > 0$
  - is a necessary condition for model validation using a **goodness-of-fit test** (χ^2)
  - We can test the if the models prediction deviates significantly from the observed data, and thus assess the quality of the model fit

Maximum Likelihood Estimation (MLE)

Fitting of MPT is done using **Maximum Likelihood Estimation (MLE)**.

- MLE does not optimize a loss or discrepancy function, but rather maximized the likelihood of the observed data given the model parameters

$$L(D|\Theta) \propto \prod_{j=1}^{J} p_j(\Theta)$$

- Note, the observed data D are fixed, the parameters Θ vary and are unknown.

- Problem:
  The likelihood is the product of the multiple probabilities for each category/data point and becomes usually very small

Log-Likelihood

The log-likelihood is the sum of the logarithms of the multiple probabilities for each category/data point

$$\log L(D|\Theta) \propto \sum_{j=1}^{J} \log p_j(\Theta)$$

Advantages:

- is easier to work with mathematically
- computationally more stable

Also often used for optimization: $\text{Deviance} = -2 \log L$

Advantages of Maximum Likelihood Estimation (MLE)

- **Consistency**: With enough data, MLE estimates converge to the true parameter values under suitable assumptions.
- **Asymptotic efficiency**: For large samples, MLEs often have the lowest possible variance among consistent estimators.
- **Interpretability**: MLE provides parameter estimates that can be interpreted in the context of the model and the data.
- **Asymptotic normality**: Their sampling distributions are often approximately normal in large samples, which supports confidence intervals and hypothesis tests.
- **Likelihood-based model comparison**: Likelihoods can be used to compare models and calculate criteria such as AIC and BIC.

Model comparison

For maximum likelihood estimates, different metrics can be calculated based on the likelihood to compare models.

- Many statistics used to assess model fit are χ^2 -distributed.
- They can then be compared using null-hypothesis significance testing (NHST).
- The χ^2 statistic is also often used as a measure of the discrepancy between observed data and model predictions (e.g., in structural equation modeling).

Model Comparison: Likelihood-Ratio Test

The likelihood-ratio test compares the **relative fit** of two nested models by taking the ratio of their likelihoods.

$$G^2 = -2 \log\left(\frac{L_0}{L_1}\right) = 2\left(\log L_1 - \log L_0\right)$$

L_0: simpler model, L_1: more complex model

- G^2 is approximately χ^2 distributed:

$$G^2 \sim \chi^2 \qquad \text{with } df = k_1 - k_0$$

k_0 and k_1 are the numbers of estimated parameters in the two models.

A **significant** result indicates that the more complex model fits significantly **better** than the simpler model. This test is typically used for nested models and relies on certain assumptions.

Model Selection: Goodness of Fit

Akaike Information Criterion (AIC)

$$\mathrm{AIC} = -2 \cdot \log L + 2 \cdot k$$

k: number of model parameters

n: sample size

Bayesian Information Criterion (BIC)

$$\mathrm{BIC} = -2 \cdot \log L + k \cdot \log(n)$$

- Like the likelihood-ratio test, AIC and BIC are used to compare models
- **Lower AIC or BIC values indicate a better-fitting model relative to the models being compared.**
- Goal: balance fit with complexity for model comparison.
- Both penalize model complexity
- BIC penalty is also dependent on sample size. With larger n, BIC would favour even simpler models.

Comparision AIC & BIC

	AIC (Akaike Information Criterion)	**BIC (Bayesian Information Criterion)**
Formula	$AIC = -2 \cdot \log L + 2k$	$BIC = -2 \cdot \log L + k \cdot \log(n)$
Penalty for parameters (k)	2 per parameter	$\log(n)$ per parameter
Philosophy	Prediction accuracy	Finding the “true” model
Sample size effect	Penalty stays constant	Penalty **increases** with n
Tends to favor	More complex models	Simpler models

Example: Retroactive Inhibition Experiment

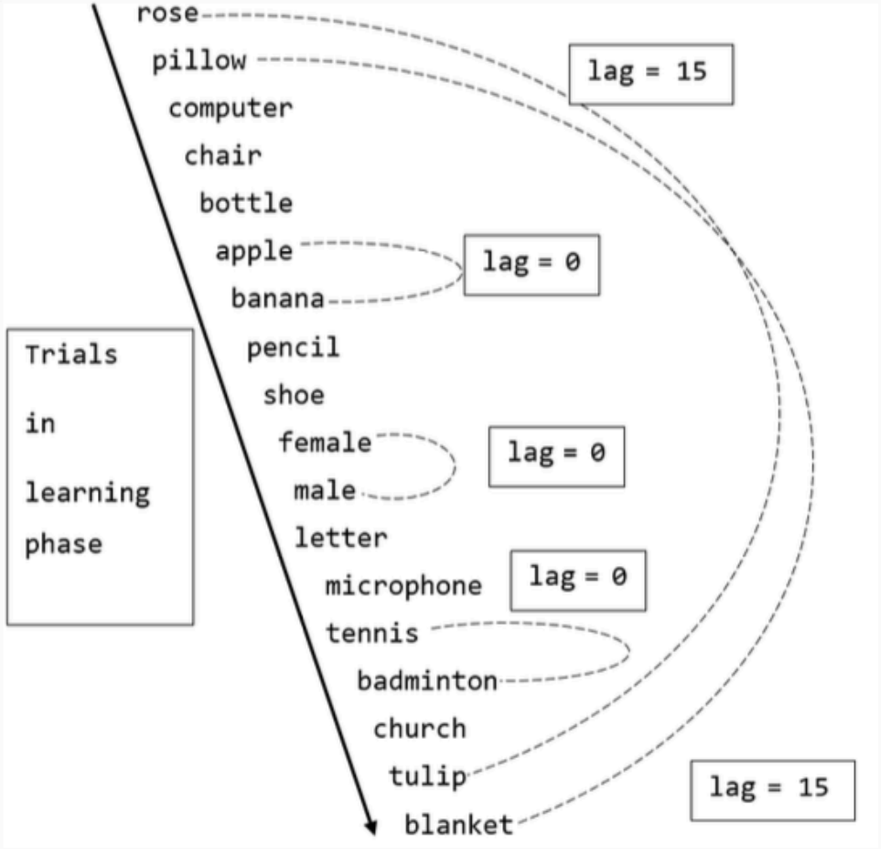
Memory Experiment by Riefer & Batchelder (1988)

classical MPT example

- Study: Lists of words
- Test: **Free recall**
- Stimuli
  - multiple semantic categories
  - 2 **associate words** per category
  - plus 5 **singletons**

Further manipulations: (*ignore for now*)

- **Age groups**: young & old participants
- **Lag**: lag 0 & lag 15 presentation

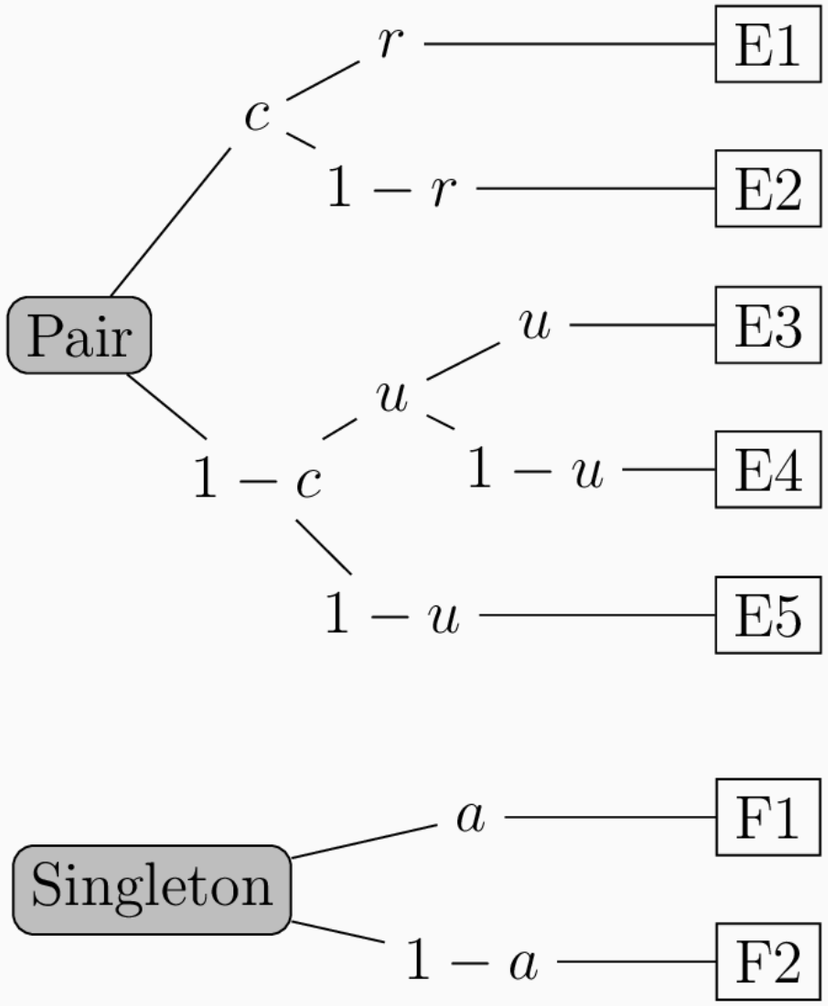


Data: Response Categories

Free recall data can be classified into six categories and pooled across young & old subjects.

- *Category pairs*
  - **E1** Pair is recalled adjacently (“aunt – uncle”)
  - **E2** Pair is recalled non-adjacently (“aunt – cow – uncle”)
  - **E3** One word in a pair is recalled
  - **E4** Neither word in a pair is recalled
- *Singletons*
  - **F1** Recall of a singleton
  - **F2** Non-recall of a singleton

Pair-Clustering Model by Riefer & Batchelder (1988)



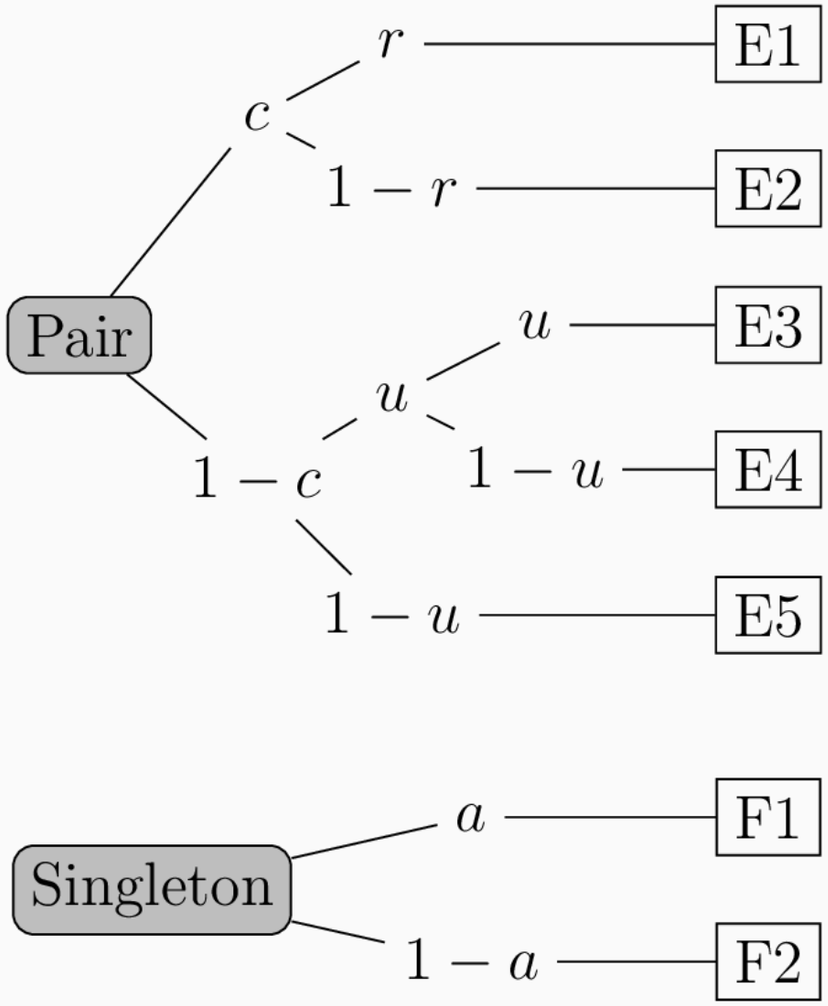
c: storing a word pair

r: retrieving a stored word pair

u: retrieving unclustered single word

a: retrieving a singleton

Pair-Clustering Model by Riefer & Batchelder (1988)



$$P(E_1|\text{Pair}) = c \cdot r$$

$$P(E_2|\text{Pair}) = (1-c) \cdot u \cdot u$$

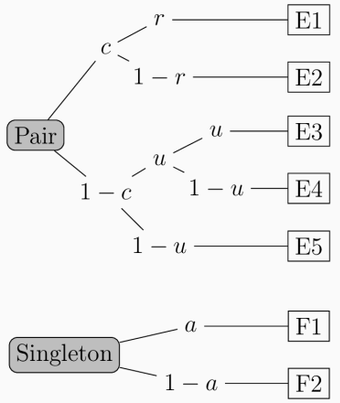
$$P(E_3|\text{Pair}) = (1-c) \cdot u \cdot (1-u) + (1-c) \cdot (1-u) \cdot u$$

$$P(E_4|\text{Pair}) = c \cdot (1-r) + (1-c) \cdot (1-u) \cdot (1-u)$$

$$P(F_1|\text{Singleton}) = a$$

$$P(F_2|\text{Singleton}) = (1-a)$$

Pair-Clustering Model by Riefer & Batchelder (1988)



1.Data

```
dat_text = "
treeid resp freq
  1     E1   132
  1     E2   19
  1     E3   147
  1     E4   502
  2     F1   166
  2     F2   634
"
dat = read.table(text=dat_text,
                 header = TRUE)
```

2. Model

```
# define model
md_pair_clst <- mptspec(
  # Pair Tree
  c * r,
  (1-c) * u * u,
  (1-c) * u * (1-u) + (1-c) * (1-u) * u,
  c * (1-r) + (1-c) * (1-u) * (1-u),
  # Singelton Tree
  a,
  (1-a)
)
```

Example

Fit model

```
md_fit1 = mpt(md_pair_clst, dat)
print(md_fit1)
```

```
Multinomial processing tree (MPT) models

Parameter estimates:
     c        r        u        a
0.4371  0.3775  0.2054  0.2075

Goodness of fit (2 log likelihood ratio):
    G2(0) = 0, p = 1
```

```
nobs(md_fit1)  ##  J = 2, number of free categories
```

```
[1] 4
```

Example: Restricted Model

Hypothesis: retrieving uncluttered single word is the same as retrieving a singleton

$$u = a$$

```
1 md_restr = update(md_pair_clst, .restr = list(u=a))
2 md_fit2 = mpt(md_restr, dat)
3 print(md_fit2)
```

```
Multinomial processing tree (MPT) models

Parameter estimates:
     c       r       a
0.4416  0.3737  0.2073

Goodness of fit (2 log likelihood ratio):
    G2(1) = 0.002445, p = 0.9606
```

Example: Restricted Model

```
1 summary(md_fit2)
```

```
Coefficients:
  Estimate Logit Estim. Std. Error z value Pr(>|z|)
c   0.4416     -0.23487    0.20365  -1.153   0.2488
r   0.3737     -0.51645    0.20258  -2.549   0.0108 *
a   0.2073     -1.34152    0.08208 -16.343   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Likelihood ratio G2: 0.002445 on 1 df, p-value: 0.9606
Pearson X2: 0.002438,    AIC: 30.597
Number of trees: 2
```

Example: Validating Restricted Model

```r
anova(md_fit2, md_fit1)  ## likelihood ratio test (order is important!)
```

```
Analysis of Deviance Table

Model 1: md_fit2
Model 2: md_fit1
  Resid. Df Resid. Dev Df  Deviance Pr(>Chi)
1         1  0.0024447
2         0  0.0000000  1 0.0024447   0.9606
```

```r
AIC(md_fit2, md_fit1)  ## AIC comparison
```

```
        df      AIC
md_fit2  3 30.59739
md_fit1  4 32.59494
```

```r
BIC(md_fit2, md_fit1)  ## BIC comparison
```

```
        df      BIC
md_fit2  3 28.75627
md_fit1  4 30.14012
```

Example: Other restrictions

Retrieving a stored word pair is the same as retrieving a singleton

$$r = a$$

```
md_restr2 = update(md_pair_clst, .restr = list(r=a))
md_fit3 = mpt(md_restr2, dat)
anova(md_fit3, md_fit1)  ## likelihood ratio test (order is important!)
```

```
Analysis of Deviance Table

Model 1: md_fit3
Model 2: md_fit1
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1         1      12.13
2         0       0.00  1    12.13 0.0004961 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example: Other restrictions

Retrieving a stored word pair is the same as retrieving a singleton

$$r = a$$

```
AIC(md_fit3, md_fit2, md_fit1)  ## AIC comparison
```

```
        df      AIC
md_fit3  3 42.72539
md_fit2  3 30.59739
md_fit1  4 32.59494
```

```
BIC(md_fit3, md_fit2, md_fit1)  ## BIC comparison
```

```
        df      BIC
md_fit3  3 40.88427
md_fit2  3 28.75627
md_fit1  4 30.14012
```

Summary Multinomial processing tree

- Multinomial processing tree (MPT) models for **categorical data**
- Responses are being described by a mixture of **latent (cognitive) processes**
- Estimating the **probabilities** of being in different latent states

- **Advantages** of MPT models
  - clear connection to psychological theory
  - conceptual and mathematical simplicity
  - easy to estimate implement in *R*

Summary: Concepts of Cognitive Modelling

- **Maximum likelihood estimation**
  - The likelihood function is the probability of the observed data as a function of the model parameters.
  - Log-likelihood
- **Degrees of freedom** depend on the number of free parameters and the number of free categories
- **Saturated models** ($df = 0$) fit the data perfectly, but cannot be validated
- Model Validation / Model Comparison
  - Nested models can be compared using likelihood-ratio tests
  - **Goodness-of-fit tests**
    - χ^2
    - AIC and BIC

Questions?